

Student Projects

Project 1: Environmental Health in a Smart Cities: Air Quality Forecasting and Optimal Sensor Placement (Submodular/Greedy + Convex)

Air pollution is an underestimated public health crisis in African cities. Rapid urbanization, traffic congestion, informal fuel usage, and unregulated industrial activities lead to dangerously high concentrations of PM_{2.5} (fine particulate matter). According to WHO estimates, air pollution contributes significantly to respiratory diseases, cardiovascular disorders, and premature mortality in Sub-Saharan Africa.

However, high-quality reference air-quality stations are extremely expensive (USD \$20,000–\$100,000 per station), and most African cities can afford only very few sensors, often placed in non-optimal locations. As a result:

- pollution hotspots remain undetected,
- exposure of vulnerable populations is underestimated,
- and public health policies are made with incomplete information.

Recent initiatives such as sensors.AFRICA deploy low-cost air-quality sensors, but even low-cost sensors are limited by:

- maintenance costs,
- power instability,
- connectivity issues,
- and sparse spatial coverage.

Therefore, cities face a fundamental optimization problem:

Where should we place a limited number of sensors, and how should we model pollution dynamics, to maximize information and public health impact?

Data

OpenAfrica sensors.AFRICA Kampala archive example:

<https://www.open.africa/bs/dataset/sensorsafrica-airquality-archive-kampala/resource/7133202b-2f4f-46e9-b5a1-3e9b8f1bc994>

Project 2: Robust Optimization of Food Distribution Networks under Demand and Transport Uncertainty in Africa

Food insecurity remains one of the most critical socio-economic challenges in Sub-Saharan Africa. Rapid population growth, climate variability, poor road infrastructure, fuel price volatility, and geopolitical instability frequently lead to food shortages, price spikes, and unequal access to basic commodities (rice, maize, millet, cassava).

In many African countries, food distribution systems are characterized by:

- a small number of central warehouses (often near ports or capitals),
- long transport routes to rural and peri-urban markets,
- unreliable road networks (seasonal road closures, flooding),
- highly uncertain demand, especially during shocks (drought, conflict, inflation).

Government agencies, NGOs, and large distributors therefore face a strategic question:

Where should we locate food warehouses and how should food flows be allocated so that shortages are minimized, costs are controlled, and the system remains robust to shocks?

Data (two sources)

- World Bank food-price-related sources (DataBank + real-time prices catalog pages):
- Road network distances via OSM (Geofabrik Africa extracts):

Geofabrik direct downloads:

<https://download.geofabrik.de/africa.html>

Example country extract (choose one region from the page, e.g., Senegal/Gambia):

<https://download.geofabrik.de/africa/senegal-and-gambia-latest.osm.pbf>

Project 3: Optimal Scheduling of Large-Scale Cloud Jobs under Energy and SLA Constraints in Africa

Across Africa, the growth of data centers, cloud platforms, and digital services (fintech, e-government, health information systems, mobile money, telecom analytics) is accelerating. However, unlike data centers in Europe or North America, African data centers face severe structural constraints:

- Unstable electricity supply (frequent outages, load shedding),
- Heavy reliance on diesel generators with high fuel costs,
- Increasing pressure to reduce carbon emissions,
- Limited computing resources compared to growing demand,
- Strict Service Level Agreements (SLAs) for critical services (banks, hospitals, telecom operators).

As a result, data center operators face a fundamental optimization challenge:

How can we schedule thousands of computational jobs in a way that minimizes energy cost and SLA violations, while respecting limited and fluctuating computing resources?

Data

Google cluster traces repository (public entry point):

<https://github.com/google/cluster-data>

Project 4: Optimal Airline Route Planning and Fleet Utilization under Economic and Infrastructure Constraints in Africa

Air transport in Africa plays a critical strategic role in economic integration, trade, tourism, healthcare access, and emergency response. However, African aviation faces unique structural challenges:

- Fragmented markets and weak regional connectivity

- High operating costs (fuel, maintenance, airport fees)
- Limited fleet sizes for regional airlines
- Underdeveloped secondary airports
- Strong competition from foreign carriers
- Political and regulatory constraints (bilateral air service agreements)

Many African cities are poorly connected, forcing passengers to route through Europe or the Middle East to travel within Africa.

As a result, regional airlines face the following question:

Which routes should we operate, with which aircraft, and at what frequency, to maximize profitability while maintaining strategic connectivity across African regions?

Data

OpenFlights route + airports data :

<https://raw.githubusercontent.com/jpatokal/openflights/master/data/airports.dat>

<https://raw.githubusercontent.com/jpatokal/openflights/master/data/routes.dat>

Project 5: Fraud and Anomaly Detection in CDR as Regularized Optimization (GD + L1 + Constraints) in Telecommunication.

Mobile telecommunications are the backbone of the African digital economy. Mobile money, voice calls, SMS, and mobile internet power:

- financial inclusion,
- e-government services,
- healthcare reporting,
- education platforms.

However, African telecom operators face large-scale fraud and abuse, including:

- SIM-box fraud (bypassing international termination fees),
- subscription fraud,
- artificial traffic inflation,
- identity misuse,
- abnormal call behavior linked to criminal activities.

These frauds cause millions of dollars in losses annually, but aggressive fraud detection creates another problem:

- too many false alarms overwhelm security teams,
- customers are wrongly blocked,
- trust in digital services is damaged.

This leads to a fundamental optimization dilemma:

How can a telecom operator detect fraud effectively while minimizing operational cost, false positives, and service disruption?

Data

Telecom CDR Fraud dataset:

<https://www.kaggle.com/datasets/mehmedumer0028/telecom-cdr-fraud-dataset>

Project 6: Cost-Sensitive Intrusion Detection and Resource-Aware Security Optimization for African Networks in cybersecurity

Across Africa, governments, banks, telecom operators, hospitals, and universities are rapidly digitizing their services. This digital transformation has dramatically increased exposure to cyberattacks, including:

- Distributed Denial-of-Service (DDoS)
- Malware and ransomware

- Port scanning and brute-force attacks
- Insider misuse
- Data exfiltration

At the same time, African organizations face severe operational constraints:

- limited cybersecurity budgets,
- small Security Operation Centers (SOCs),
- shortage of skilled analysts,
- unreliable network infrastructure.

As a result, African SOC face a critical dilemma:

If we detect too little, attacks succeed. If we detect too much, analysts are overwhelmed and systems are disrupted.

Data

CICIDS2017 official dataset page:

<https://www.unb.ca/cic/datasets/ids-2017.html>

Project 7: Robust Portfolio Optimization for African Financial Markets **under Risk, Liquidity, and Market Uncertainty in finance.**

African financial markets are growing rapidly, driven by:

- mobile money and fintech adoption,
- expanding stock exchanges (Johannesburg, Lagos, Nairobi, Casablanca),
- increased participation from retail and institutional investors.

However, these markets exhibit structural characteristics that strongly differentiate them from developed markets:

- high volatility and regime shifts,
- low liquidity for many assets,
- currency instability and inflation risk,
- sudden market closures or trading halts,
- strong sector concentration (banking, energy, telecom).

As a result, classical portfolio theory assumptions (Gaussian returns, stable covariance, perfect liquidity) are often violated.

This leads asset managers and fintech platforms to a fundamental question:

How can we optimally allocate capital across African assets while controlling risk, liquidity exposure, and robustness to market shocks?

Data

Africa Stock Dataset:

<https://www.kaggle.com/datasets/ogunmusireseyi/africa-stock-dataset>

Project 8: Optimal Intervention Policy for an Epidemic (Gradient Descent + Optimal Control)

African financial markets are characterized by structural fragilities that make classical portfolio theory dangerously misleading if applied naïvely.

Key structural realities include:

- Thin trading volumes: many assets trade irregularly → unreliable prices
- Liquidity traps: investors cannot exit positions quickly without price impact
- Macroeconomic instability: inflation, currency devaluation, political shocks
- Market segmentation: poor cross-country correlation stability

- Regulatory constraints: pension funds and institutional investors face strict allocation limits

As a consequence, an “optimal” portfolio on paper may be:

- impossible to rebalance,
- extremely risky in downturns,
- or illegal under regulation.

Therefore, portfolio construction in Africa is fundamentally an optimization-under-constraints and uncertainty problem.

Data

Use OWID COVID dataset

<https://covid.ourworldindata.org/data/owid-covid-data.csv>

Minimum deliverables:

1. Problem statement (Africa context + stakeholders + constraints+ objective)
2. Mathematical formulation (variables, objective, constraints)
3. Solution approach (algorithm + complexity + why chosen)
4. Python notebook (reproducible, clear plots, final results)
5. 15-20 slide Beamer (storytelling + visuals)